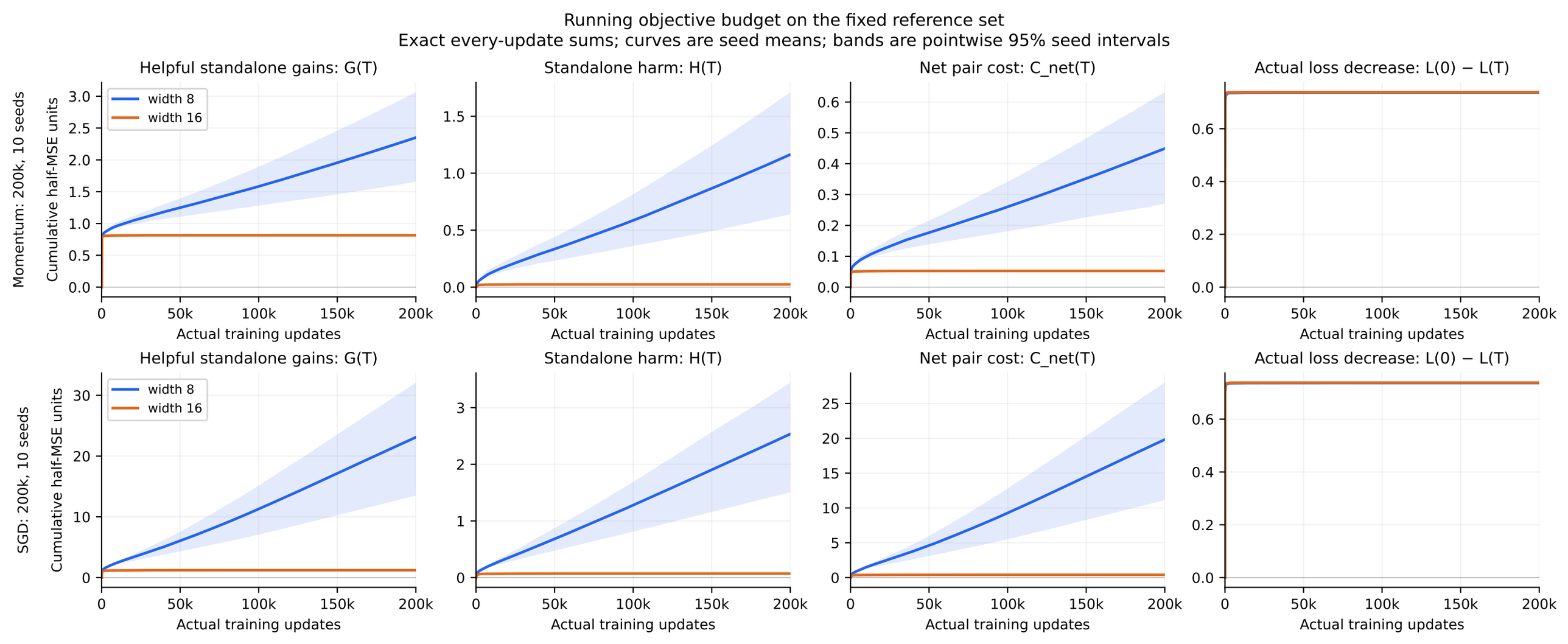

Running objective budget on the fixed reference set
Exact every-update sums; curves are seed means; bands are pointwise 95% seed intervals
Momentum: 200k, 10 seeds
SGD: 200k, 10 seeds
Helpful standalone gains: G(T)
Standalone harm: H(T)
Net pair cost: C_net(T)
Actual loss decrease: L(0) − L(T)
Cumulative half-MSE units
Actual training updates
width 8
width 16
0
50k
100k
150k
200k